

Physik II — FS 2024 — Prof. R. Wallny — Serie 6

Solution

Exercise 1: Two conducting spheres

First, we note that electrostatic induction between the two spheres is negligible, because they are far away from each other, $d \gg R_1, R_2$.

Before the connection, the electrostatic potentials of the two spherical surfaces are given by

$$\phi_1 = \frac{Q_1}{4\pi\epsilon_0 R_1}, \quad \phi_2 = \frac{Q_2}{4\pi\epsilon_0 R_2}. \quad (1)$$

The connection allows the flow of charge from one sphere to the other: the (positive) charge flows from the larger to the smaller potential, until both spheres are at the same potential. This equilibrium is described by the new charges Q'_1 and Q'_2 and by the new common potential

$$\phi' = \frac{Q'_1}{4\pi\epsilon_0 R_1} = \frac{Q'_2}{4\pi\epsilon_0 R_2}. \quad (2)$$

Moreover the charge is conserved:

$$Q_1 + Q_2 = Q'_1 + Q'_2. \quad (3)$$

Solution of equations (2) und (3) gives the new charges

$$Q'_1 = \frac{R_1(Q_1 + Q_2)}{R_1 + R_2}, \quad Q'_2 = \frac{R_2(Q_1 + Q_2)}{R_1 + R_2}, \quad (4)$$

and the new potential

$$\phi' = \frac{Q_1 + Q_2}{4\pi\epsilon_0(R_1 + R_2)}. \quad (5)$$

With Gauss's Theorem, applied at the spherical surfaces (far away from each other), we can determine the electric fields. Due to symmetry reasons, the electric field close to each sphere is pointing in radial direction and its absolute value is given by

$$4\pi R_i^2 \cdot E'_i = \frac{1}{\epsilon_0} Q'_i \quad \Rightarrow \quad E'_i = \frac{Q'_i}{4\pi\epsilon_0 R_i^2}. \quad (6)$$

The electric fields at the spherical surfaces are

$$E'_1 = \frac{Q_1 + Q_2}{4\pi\epsilon_0(R_1 + R_2)R_1} = \frac{\phi'}{R_1}, \quad E'_2 = \frac{Q_1 + Q_2}{4\pi\epsilon_0(R_1 + R_2)R_2} = \frac{\phi'}{R_2}. \quad (7)$$

Exercise 2: Cylindrical capacitor

Let us use Gauss's law for the calculation of the field. The integration surface is a cylinder of length h and radius r . The z and ϕ coordinates vanish, thanks to the symmetry of the system. Then,

$$\vec{E} \cdot d\vec{S} = E(r)d\Omega = E(r)r d\phi dz.$$

Let us now consider the different regions in turn:

- a) • Region a:

$$\iint_S \vec{E} \cdot d\vec{S} = \frac{1}{\epsilon_0} Q_{\text{in}} = 0 \Rightarrow E_a \equiv 0$$

- Region b:

$$\iint_S \vec{E} \cdot d\vec{S} = 2\pi r h E(r) = \frac{1}{\epsilon_0} Q_i \Rightarrow E_b(r) = \frac{Q_i}{\epsilon_0 2\pi r h}$$

- Region c:

$$\iint_S \vec{E} \cdot d\vec{S} = 2\pi r h E(r) = \frac{1}{\epsilon_0} (Q_i + Q_a) \Rightarrow E_c(r) = \frac{Q_i + Q_a}{\epsilon_0 2\pi r h}$$

- b) The charge spreads on the different surfaces as follows:

- Inside of the inner cylinder: $Q = 0$
- Outside of the inner cylinder: $Q = Q_i$
- Inside of the outside cylinder: $Q = -Q_i$ in order to have a vanishing field within the outside cylinder.
- Outside of the outside cylinder: $Q = Q_i + Q_a$

- c) In order to define a capacitance, we need a charge and a potential. The relevant charge Q is that on the inner cylindrical shell Q_i . Due to Gauss' Law and the superposition principle, the charge on the outer shell does not have an influence on the electric field in region b and does not play a role for the capacitance. The potential difference is calculated in cylindrical coordinates as

$$\begin{aligned} V &= \int_{r_i}^{r_a} \vec{E}(\vec{r}) \cdot d\vec{r} \\ &= \int_{r_i}^{r_a} \frac{Q_i}{\epsilon_0 2\pi r h} dr \\ &= \frac{Q_i}{\epsilon_0 2\pi h} \ln \frac{r_a}{r_i} \end{aligned}$$

Thus the capacitance is $C = \frac{Q_i}{V} = 2\pi h \epsilon_0 \ln \frac{r_a}{r_i}$.

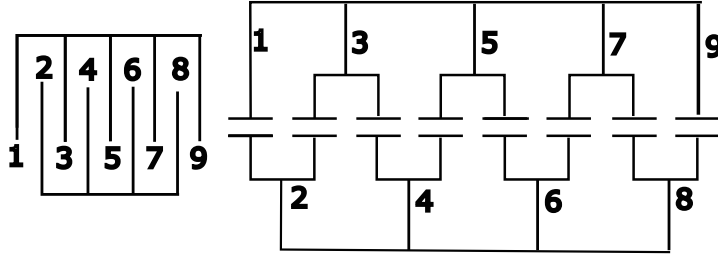
Exercise 3: Rotary Variable Capacitor

- a) The area overlap of the two capacitor plates is given by

$$A = \frac{\pi R^2}{2} - \pi R^2 \frac{\phi}{360} \quad (8)$$

The capacity is given by

$$C = \frac{\epsilon_0 A}{d} \quad (9)$$



- b) A variable capacitor is equivalent to a parallel connection of plate capacitors. All rotatable plates are short-circuited and all static plates are short-circuited (see figure). The total capacity is, therefore, by the sum of the single capacities. An arrangement of nine plates is equivalent to the parallel connection of eight capacitors. The maximum capacity is reached when $\phi = 0$. The overlap is then a semicircle.

This leads to

$$C_{max} = \frac{\epsilon_0 A_{ges}}{d} = \frac{8\epsilon_0 \pi R^2}{2d} \quad (10)$$

Insertion of the given parameters gives $C_{max} = 2 \text{ nF}$.

The energy stored by a capacitor is given by

$$E_{el} = \frac{1}{2} CV^2 \quad (11)$$

With $V = 3 \text{ V}$ it follows $E_{el} = 9 \text{ nJ}$.

- c) The total capacity of a series circuit of capacitors is given by

$$C_{ges} = \frac{C_1 C_2}{C_1 + C_2} \quad (12)$$

With the result from the previous task $C_{ges} = 1 \text{ nF}$.

Exercise 4: Does the shape play a role?

- a) Let A be a closed area, that fully encloses the positively charged electrode. The current I out of this electrode can be described as the flow of the current density $\vec{j}$ through the area A :

$$I = \int_A \vec{j} \cdot \vec{ds}. \quad (13)$$

With the (microscopic) ohmic law $\vec{j} = \sigma \vec{E}$, we get

$$I = \int_A \vec{j} \cdot \vec{ds} = \sigma \int_A \vec{E} \cdot \vec{ds}. \quad (14)$$

On the right side of the equation, we recognize the flow of the electric field through A and with the help of Gauss's law,

$$\int_A \vec{E} \cdot d\vec{s} = \frac{Q}{\varepsilon_0}, \quad (15)$$

we can write

$$I = \int_A \vec{j} \cdot d\vec{s} = \sigma \int_A \vec{E} \cdot d\vec{s} = \frac{\sigma Q}{\varepsilon_0}, \quad (16)$$

where Q is the charge on the electrode. This charge is connected to the potential difference between the electrodes V through the equation $Q = C \cdot V$ and it follows:

$$I = \int_A \vec{j} \cdot d\vec{s} = \sigma \int_A \vec{E} \cdot d\vec{s} = \frac{\sigma Q}{\varepsilon_0} = \frac{\sigma CV}{\varepsilon_0}. \quad (17)$$

The resistance between the electrodes is therefore

$$R = \frac{V}{I} = \frac{\varepsilon_0}{\sigma C}, \quad (18)$$

which was to be proven.

- b) The resistance in equation (18) does indeed depend on the shape of the geometry of the system. But this information is encoded in the value of the capacitance C . The capacitance does depend on the geometry and material of the system.
- c) Let $Q(t)$ be the charge at time t on the positively charged electrode. The charge is related to the current between the electrodes through

$$I = -\frac{dQ}{dt}. \quad (19)$$

Also, we know from equation (16) from part a) that $I = \sigma Q/\varepsilon_0$. From this we get the differential equation for the charge Q :

$$\frac{dQ}{dt} = -\frac{\sigma}{\varepsilon_0} Q. \quad (20)$$

The solution is

$$Q(t) = B e^{-t/\tau}, \quad (21)$$

with a constant B and with

$$\tau = \frac{\varepsilon_0}{\sigma}. \quad (22)$$

We have found that the electrodes discharge exponentially in time. The constant B is given by the initial condition $Q(t=0) = Q_0$. Therefore, $B = Q_0$ and the final result is

$$Q(t) = Q_0 e^{-t/\tau}. \quad (23)$$

Exercise 5: Current through cube

- a) The temporal change of the charge density can be determined through the continuity equation:

$$\frac{d\rho}{dt} = -\vec{\nabla} \cdot \vec{j} = (-2x + 4y - 3) \text{ mA cm}^{-3} \quad (24)$$

b) The current through a closed surface is equal to the change of charge density inside the volume:

$$I_{\partial V} = - \int_V \frac{d\rho}{dt} dV \quad (25)$$

$$= \int_{-1 \text{ cm}}^{1 \text{ cm}} \int_{-1 \text{ cm}}^{1 \text{ cm}} \int_{-1 \text{ cm}}^{1 \text{ cm}} (2x - 4y + 3) dx dy dz \quad (26)$$

$$= \int_{-1 \text{ cm}}^{1 \text{ cm}} \int_{-1 \text{ cm}}^{1 \text{ cm}} (-8y + 6) dy dz \quad (27)$$

$$= \int_{-1 \text{ cm}}^{1 \text{ cm}} 12 dz \quad (28)$$

$$= 24 \text{ mA} \quad (29)$$

$$(30)$$

c) Due to the minus sign in equation 25, $I_{\partial V}$ is the current that flows out of the volume. After $\Delta t = 10 \text{ s}$ the charge inside the cube is therefore:

$$Q_V = -I_{\partial V} \cdot \Delta t \quad (31)$$

$$= -24 \text{ mA} \cdot 10 \text{ s} \quad (32)$$

$$= -0.24 \text{ C} \quad (33)$$